

Wide-Angle Multislice Propagation for Electron Microscopy: Comparison of Fresnel, Angular-Spectrum, and Wave-Propagation Methods

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Abstract

Conventional Fresnel multislice (F-MS) is the standard method for modeling electron-sample interactions in transmission electron microscopy (TEM). It is computationally efficient and can simulate both perfect and defective crystals, but its paraxial propagation kernel loses accuracy when large scattering angles contribute to the exit wave. Here we compare slice-based FFT propagation methods that preserve the practical multislice workflow while improving wide-angle propagation. We introduce wave-propagation multislice (WP-MS), an adaptation of the wave propagation method from scalar optics, and compare it with F-MS and angular-spectrum multislice (AS-MS). The three methods are first benchmarked against a direct ordinary-differential-equation (ODE) solution of the second-order Klein-Gordon Helmholtz equation for a representative Au [100] sample. We then compare their predictions for convergent-beam electron diffraction (CBED) in Si [111] and Au [100]. AS-MS removes most of the wide-angle error of F-MS at essentially the same per-slice computational cost. WP-MS is closest to the ODE reference, but its additional accuracy over AS-MS is modest under the high-energy conditions studied here.

Keywords: electron microscopy, multislice, electron diffraction, TEM image simulation, Fresnel propagator, angular spectrum, wave propagation method

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1. Introduction

The traditional image simulation workflow in electron microscopy assumes a structural model, calculates an image from that model, and qualitatively compares the agreement with experiment via trial and error. In modern reconstruction workflows, including ptychography and related phase-retrieval methods, the recorded experimental data are now used to recover a projected potential, phase shift, or three-dimensional structure, where the electron-matter interaction model is not merely a tool for visual comparison after the experiment, but part of the reconstruction algorithm itself. The accuracy of the forward model is therefore of critical importance to the accuracy of the recovered structure.

At the most fundamental level, electron propagation through matter is a time-dependent many-body quantum problem. The incident electron interacts with the specimen electrons and nuclei, so a complete description would include elastic and inelastic scattering channels, multiple scattering, possible energy exchange with the specimen, and in principle both forward- and backward-propagating components. Such a direct treatment is not practical for realistic TEM specimens, and image simulations instead rely on reduced stationary wave equations. Multislice methods provide such a practical numerical tool to simulate the electron-specimen interaction.

The standard implementation is paraxial Fresnel multislice (F-MS) [1, 2]. F-MS is simple, fast, and well suited to conventional high-energy TEM and STEM simulations of thin, low- Z specimens. In many experimental comparisons, uncertainties such as thermal diffuse scattering, detector response, specimen drift, or imperfect aberration calibration dominate, so F-MS is often adequate. However, for low incident energies, thick specimens, high- Z materials, or simulations that retain high-angle diffraction information, F-MS can introduce significant errors because its propagation kernel is paraxial and because standard forward multislice neglects backscattering [3–6].

The smallest correction to F-MS is to replace the parabolic Fresnel propagator between slices by the exact one-way spherical-wave propagator. As far as we are aware, Ming et al. [6] were the first to explicitly propose the full spherical propagator correction in electron microscopy. We refer to this method as angular-spectrum multislice (AS-MS), following the terminology used in scalar light optics [7]. AS-MS keeps the same split-step structure and the same FFT-based cost as F-MS, but restores the exact free-space dispersion relation. It is therefore a minimal wide-angle correction, although it

still treats the specimen as a sequence of projected phase gratings separated by vacuum propagation.

More complete wide-angle methods seek to improve the propagation inside the specimen, or to use the full second-order wave equation as a reference. The Chen–Van Dyck method, also known as fully corrected multislice (FC-MS), expands the square-root propagation operator in a Taylor series and retains higher-order terms that couple the specimen potential to transverse derivatives [6, 8]. This improves the treatment of large scattering angles, low-energy electrons, thick specimens, and high- Z materials, and in its more complete form gives access to backward-propagating components. The cost is that the square-root operator is no longer a simple Fourier-space phase factor, and can be expensive to compute depending of the number of terms chosen to solve in the series expansion.

A closely related method to that presented by Chen–Van Dyck was studied earlier in scalar light optics, and are known as Padé beam-propagation methods, which approximate the same square-root operator by rational functions rather than Taylor polynomials [9]. These approximants can represent the square-root dispersion relation over a wider angular range than a truncated Taylor series of the same order, but they still introduce substantial overhead relative to F-MS or AS-MS. Finally, direct integration of the full second-order wave equation as an ordinary differential equation (ODE) provides a more accurate numerical reference, and has been used to assess paraxial and corrected multislice methods [3, 5]. These studies indicate that a dominant high-angle error of F-MS arises from the parabolic propagation step. Hillier et al. [4] also showed that retaining the second longitudinal derivative is important for the positions of Laue-zone rings in low-energy Si CBED patterns. However, ODE and higher-order corrected methods are too expensive to use directly as routine forward models in large-scale inverse reconstructions.

In order to overcome the issue of needing an accurate and efficient solution to the Time Independent wave equation, the light-optics community has developed alternative methods that retain the efficiency of the FFT while avoiding the need to choose a Taylor or Padé expansion order. One such method is the wave propagation method (WPM), introduced by Brenner and Singer [10]. WPM treats the medium as locally homogeneous over small transverse regions, applies homogeneous angular-spectrum propagators for representative refractive indices, and recombines the resulting fields according to the local refractive-index distribution. This retains a potential-dependent

wavelength within the propagation step without explicitly forming the full dense square-root operator. Efficient and differentiable variants have been developed in scalar optics literature [11, 12]. The method is therefore a natural starting point for a practical wide-angle multislice algorithm in electron microscopy, but to our knowledge it has not yet been systematically assessed in this context.

Table 1 summarizes the related developments in light and electron optics for the method families used or compared here.

Method family	Light-optics reference	Electron-optics reference	Description
Fresnel MS	Fresnel (1818); Fleck–Morris–Feit (1976) [13, 14]	Cowley–Moodie (1957); Ishizuka–Uyeda (1977) [1, 2]	Split-step propagation with the paraxial Fresnel kernel.
Angular Spectrum MS	Booker–Clemmow (1950); Korpel et al. (1986) [15, 16]	Ming–Chen (2013) [6]	Same split-step structure, but with the exact free-space propagator.
Padé/Series Expansion (BPMs)	Hadley (1992); Lee–Voges (1994) [9, 17]	Chen–Van Dyck (1997) [8]	Higher-order series expansion approximations to the square-root propagation operator.
Wave Propagation Method	Brenner–Singer (1993); Schmidt et al. (2016) [10, 11]	This work	Local homogeneous angular-spectrum kernels applied inside the medium.
ODE	Banerjee–Sharma (1989) [18, 19]	Rother–Scheerschmidt (2009)*; Wacker–Schröder (2015) [3, 5]	Reciprocal-space and direct references for non-paraxial propagation.

*Rother–Scheerschmidt used a forward-scattering approximation in Fourier space for a periodic crystal, rather than direct real-space integration of the full second-order ODE.

Table 1: Historical correspondence between the propagation methods used here and closely related developments in light and electron optics.

The aim of this work is to adapt the WPM from the light optics community (which we call Wave Propagation Multislice, WP-MS) and assess it’s benefits in the context of electron microscopy. We therefore compare F-

MS, AS-MS, and the wave-propagation multislice method (WP-MS) against a forward ODE reference solution of the full second-order Klein–Gordon Helmholtz equation, and then test the multislice methods against each other in representative Si [111] and Au [100] CBED simulations. The comparison identifies when F-MS remains sufficient, when the simple AS-MS correction captures the dominant error, and if the local-wavelength correction in WP-MS becomes relevant.

2. Wave-Equation Models and Propagation Algorithms

2.1. Wave equation and refractive index

Throughout this work we use spatial frequencies in cycles per unit length, rather than angular wavenumbers in radians per unit length. The vacuum spatial frequency is therefore

$$k_0 = \frac{1}{\lambda}, \quad (1)$$

where λ is the relativistic electron wavelength. We use the Fourier transform convention

$$\mathcal{F}\{f(x, y)\}(k_x, k_y) = \int f(x, y) \exp[-2\pi i(k_x x + k_y y)] dx dy, \quad (2)$$

and write $k_\perp^2 = k_x^2 + k_y^2$.

For high-energy elastic scattering in an electrostatic potential, we start from the scalar relativistically corrected Schrödinger equation [20, 21]

$$\left[-\frac{\hbar^2}{2\gamma m_0} \nabla^2 + E_{\text{pot}}(\mathbf{r}) \right] \psi(\mathbf{r}) = \frac{p_0^2}{2\gamma m_0} \psi(\mathbf{r}), \quad (3)$$

where ψ is the spatial wavefunction, m_0 is the electron rest mass, γ is the Lorentz factor, and $p_0 = h/\lambda = \hbar k_0$ is the vacuum momentum. We write the electrostatic potential in volts as $V(\mathbf{r})$ and the corresponding electron potential energy as $E_{\text{pot}}(\mathbf{r}) = -eV(\mathbf{r})$, with $e > 0$.

Multiplying equation (3) by $-2\gamma m_0/\hbar^2$ gives a Helmholtz equation for the same wavefunction,

$$\nabla^2 \psi + (2\pi k_0)^2 n_S^2(\mathbf{r}) \psi = 0, \quad (4)$$

where the Schrödinger refractive index is

$$n_S^2(\mathbf{r}) = 1 - \frac{2\gamma m_0}{\hbar^2 k_0^2} E_{\text{pot}}(\mathbf{r}) = 1 + \frac{\sigma}{\pi k_0} V(\mathbf{r}). \quad (5)$$

and the conventional TEM interaction constant is

$$\sigma = \frac{2\pi\gamma m_0 e}{h^2 k_0} = \frac{2\pi\gamma m_0 e \lambda}{h^2}. \quad (6)$$

The identification of the electrostatic potential with an electron-optical refractive index is due to [22–24]. The refractive index notation is useful because the choice of scalar wave equation can be expressed through the choice of $n(\mathbf{r})$, separating approximations to the wave equation from approximations to the potential. For comparison, the time-independent Klein–Gordon equation gives

$$n_{\text{KG}}^2(\mathbf{r}) = \frac{[E_{\text{tot}} - E_{\text{pot}}(\mathbf{r})]^2 - m_0^2 c^4}{E_{\text{tot}}^2 - m_0^2 c^4}. \quad (7)$$

Here E_{tot} is the incident-electron energy in vacuum including the rest energy, so for an electron accelerated through a voltage U , $E_{\text{tot}} = m_0 c^2 + eU = \gamma m_0 c^2$.

The relativistically corrected Schrödinger refractive index is a linearised potential approximation to the Klein–Gordon refractive index, while retaining the relativistic vacuum momentum through k_0 . Therefore changing from n_S to n_{KG} changes the potential model, whereas the choice between Fresnel, angular-spectrum, and WP-MS propagation changes how a given input field through a refractive-index distribution is propagated.

2.2. Fresnel and angular-spectrum multislice

To connect equation (4) to conventional multislice, we make the standard high-energy, or slowly varying envelope approximation used in electron microscopy [21]. Writing the wavefunction as a rapidly oscillating carrier multiplied by a slowly varying envelope,

$$\psi(x, y, z) = \exp(2\pi i k_0 z) \tilde{\psi}(x, y, z), \quad (8)$$

gives the parabolic wave equation

$$\frac{\partial \tilde{\psi}}{\partial z} = \frac{i}{4\pi k_0} \nabla_{\perp}^2 \tilde{\psi} + i\pi k_0 [n^2(x, y, z) - 1] \tilde{\psi}. \quad (9)$$

For the corrected Schrödinger model, the interaction term reduces to $i\sigma V \tilde{\psi}$. For slice j , define the projected electrostatic potential and transmission function by

$$V_j(x, y) = \int_{z_j}^{z_j + \Delta z} V(x, y, z) dz, \quad t_j(x, y) = \exp[i\sigma V_j(x, y)]. \quad (10)$$

Fresnel multislice (F-MS) applies this transmission function and then propagates the wave through vacuum with the paraxial Fresnel kernel. Dropping the tilde for notational simplicity, one slice can be written as

$$\psi_{j+1}(x, y) \approx \mathcal{F}^{-1} \left[\exp \left(-i\pi \frac{k_{\perp}^2}{k_0} \Delta z \right) \mathcal{F} \{ t_j(x, y) \psi_j(x, y) \} \right]. \quad (11)$$

Computing the above equation is fast because each slice requires only one FFT pair. Its approximation is that the free-space longitudinal wavevector has been expanded as

$$k_z = \sqrt{k_0^2 - k_{\perp}^2} \simeq k_0 - \frac{k_{\perp}^2}{2k_0}.$$

After removal of the carrier phase $\exp(2\pi i k_0 z)$, this expansion gives the Fresnel kernel in equation (11). Conventional multislice is therefore both a split-step approximation to the specimen and a small-angle approximation to the longitudinal dispersion relation.

The angular spectrum method keeps the same split-step structure but replaces the Fresnel kernel by the exact free-space dispersion relation:

$$\psi_{j+1}(x, y) \approx \mathcal{F}^{-1} \left[\exp \left(2\pi i \left[\sqrt{k_0^2 - k_{\perp}^2} - k_0 \right] \Delta z \right) \mathcal{F} \{ t_j(x, y) \psi_j(x, y) \} \right]. \quad (12)$$

Apart from the change in the Fourier-space phase factor, the algorithm is the same as standard multislice and has the same $\mathcal{O}(P \log P)$ per-slice cost, where P is the number of transverse pixels. It is therefore the smallest change one can make to F-MS in order to restore the exact free-space dispersion relation. The approximation that remains is the split-step treatment of the specimen: the potential is still applied as a thin transmission function, while propagation between slices is still performed in vacuum.

2.3. Classical Wave Propagation Method in scalar optics

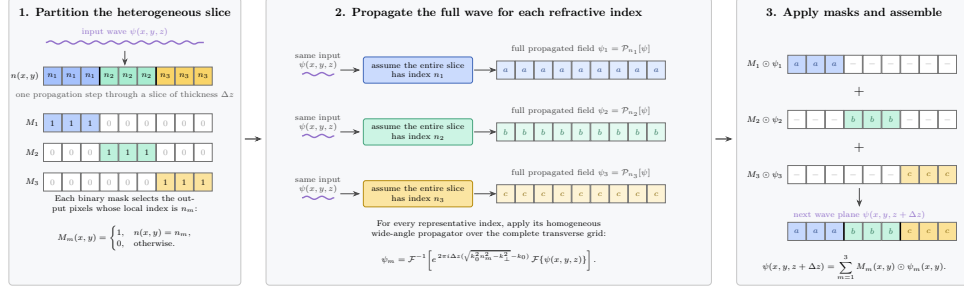


Figure 1: Schematic of the Wave Propagation Method (WPM) as presented by [11]. The input wave is propagated using a bank of homogeneous angular-spectrum kernels, each evaluated at a different representative refractive index n_m . At each output pixel, the propagated field corresponding to the local refractive index is selected via mask and recombined. Our improvement to this algorithm (not described in this figure for simplicity) includes a soft binning step which makes this method suitable for a continuous refractive index distribution - see Figure 2 below.

The Wave Propagation Method (WPM) addresses the remaining vacuum-propagation approximation in AS-MS by allowing the propagation kernel itself to depend on the refractive index of the medium. In this sense, the local interaction with the medium is incorporated into a single propagation update, rather than being applied as a separate transmission function. Returning to the full second-order Helmholtz equation (4), it can be rearranged into an operator equation with a forward and backward branch [11]. Ignoring the backward branch, the forward branch may be written as

$$\partial_z \psi = 2\pi i \left[\sqrt{k_0^2 n^2(x, y, z) + \frac{1}{4\pi^2} \nabla_\perp^2} - k_0 \right] \psi. \quad (13)$$

The central concern is that the square-root operator inside the wave equation mixes the potential and the transverse derivative, and cannot be evaluated as a single Fourier-space propagator. Padé BPM and FC-MS methods solve for this operator using rational or series approximations to the square-root term, which are expensive and cannot usually directly use the FFT.

Looking again at the AS-MS propagation kernel of equation (12),

$$\left[\sqrt{k_0^2 - k_\perp^2} - k_0 \right]$$

what is primarily missing is a refractive-index term. We can reintroduce this term locally by replacing k_0 with $k_0 n(x, y)$, giving

$$\left[\sqrt{k_0^2 n^2(x, y) - k_\perp^2} - k_0 \right].$$

For a single refractive-index value, this kernel can be evaluated straightforwardly with the angular spectrum method. The traditional WPM as introduced by Brenner simply extends the angular spectrum method by evaluating a bank of homogeneous angular-spectrum propagators for each refractive index in the slice, and then retaining the propagated field corresponding to the local refractive index at each output pixel.

This operation is obviously simplistic, because output pixels can receive a propagated field that travelled from some pixels with the incorrect refractive index. However, the approximation lives in the fact that those different refractive indices are further away from the local refractive-index value of that pixel, and therefore contribute less to the final result. In this manner, the propagation step accounts for the local wavelength inside the specimen without explicitly evaluating the full square-root operator. Schmidt et al. [11] provided a more rigorous derivation motivating the WPM, and showed that the leading error from neglecting coupling between neighbouring refractive-index regions affects amplitude rather than phase; consequently, the WPM phase propagation is correct up to second order.

Figure 1 shows the same local-homogeneous idea in the form used by our electron-microscopy adaptation below. In the classical WPM as implemented by Schmidt, a mask is created to separate out each unique refractive index in a slice, the angular spectrum kernel is calculated for each refractive index, and then the mask is used to choose the correct angular spectrum solution from the bank of all computed propagations. Schmidt et al. [11] made the method practical for many optical simulations by grouping pixels with the same refractive index, as only one homogeneous angular-spectrum propagation is then required for each distinct index value. If a slice contains M such values, the cost is reduced to approximately

$$\mathcal{O}(MP \log P),$$

instead of a direct all-to-all evaluation.

This saving is large in light-optical simulations where a slice may contain only a small number of unique refractive indices. It is less immediate

in electron microscopy, where the electrostatic potential varies continuously and almost every pixel may have a different refractive index. Exact grouping therefore gives little benefit unless the refractive-index field is deliberately discretised. A second issue is that hard grouping introduces sharp masks, which are inconvenient for gradient-based optimisation because the assignment of pixels to refractive-index groups is not smoothly differentiable.

2.4. Wave propagation multislice

We refer to our electron-microscopy adaptation of WPM as wave propagation multislice (WP-MS), to distinguish it from the classical WPM used in scalar light optics. The main practical issue is that an electron-microscopy slice has a nearly continuous refractive-index distribution, so grouping exactly by refractive-index value would require almost one homogeneous propagation per pixel. We therefore use a differentiable binned version of WPM, partly following the soft-partition idea of [12], but generalised from two refractive-index values to many reference values.

For each slice, we first discretise the refractive-index range $[n_{\min}, n_{\max}]$ into N_{bins} discrete reference values, denoted as $\{n_{\text{ref}}^{(k)}\}_{k=1}^{N_{\text{bins}}}$.

Most pixels in a typical slice are close to vacuum, with $n \simeq 1$, whereas the larger refractive-index values associated with atoms occupy a smaller fraction of the slice. We therefore use non-uniform reference values rather than linearly spaced bins, allocating more bins near $n = 1$. The reference indices are distributed according to

$$n_{\text{ref}}^{(k)} = n_{\min} + (n_{\max} - n_{\min}) \left(\frac{k-1}{N_{\text{bins}}-1} \right)^\gamma, \quad k = 1, \dots, N_{\text{bins}}, \quad (14)$$

with $\gamma = 2$ in the simulations below. Values $\gamma > 1$ give denser binning near the vacuum refractive index. See Figure 2 for a visualisation of the binning algorithm.

Next, for a propagation step Δz , we compute a bank of propagated fields $u_k(x, y)$. Each field u_k corresponds to the diffraction of the input field $u_{\text{in}}(x, y)$ assuming a homogeneous medium with refractive index $n_{\text{ref}}^{(k)}$. This is performed efficiently using batch FFT operations with the angular spectrum kernel:

$$u_k(x, y) = \mathcal{F}^{-1} \left[\mathcal{F}\{u_{\text{in}}\} \exp \left(2\pi i \Delta z \left[\sqrt{(k_0 n_{\text{ref}}^{(k)})^2 - k_x^2 - k_y^2} - k_0 \right] \right) \right]. \quad (15)$$

To evaluate the field at a specific spatial coordinate (x, y) where the local refractive index is $n(x, y)$, we identify the two bounding reference bins such that $n_L \leq n(x, y) < n_R$, where $n_L, n_R \in \{n_{\text{ref}}^{(k)}\}$. We adopt a differentiable interpolation scheme by defining a local interpolation weight

$$w_{\text{raw}} = \frac{n(x, y) - n_L}{n_R - n_L}, \quad w_{\text{raw}} \in [0, 1]. \quad (16)$$

To obtain smooth gradients at the bin boundaries, we map this linear weight through the cubic smoothstep function

$$w = S(w_{\text{raw}}) = 3w_{\text{raw}}^2 - 2w_{\text{raw}}^3. \quad (17)$$

Linear interpolation is already differentiable inside each bin, but its gradient is discontinuous at the bin boundaries.

The final field at the next step $u(x, y, z + \Delta z)$ is computed as the weighted superposition of the fields propagated at the bounding indices:

$$u(x, y) = (1 - w) \cdot u_L(x, y) + w \cdot u_R(x, y). \quad (18)$$

The local refractive index first determines the two bounding reference indices n_L and n_R , and the weight w specifies the fractional position of $n(x, y)$ between those two values after the smoothstep mapping. The fields u_L and u_R are the full propagated fields obtained by assuming homogeneous media with refractive indices n_L and n_R , respectively, and the output pixel is formed from their local two-bin mixture.

Equivalently, the WP-MS update can be written as

$$\begin{aligned} \psi(x, y, z + \Delta z) \approx \sum_{m=1}^{N_{\text{bins}}} w_m(x, y) \mathcal{F}^{-1} \left[\exp \left(2\pi i \left[\sqrt{k_0^2 (n_{\text{ref}}^{(m)})^2 - k_{\perp}^2} - k_0 \right] \Delta z \right) \right. \\ \left. \times \mathcal{F} \{ \psi(x, y, z) \} \right]. \end{aligned} \quad (19)$$

This gives a differentiable approximation to the one-way medium propagator in equation (13). When the slice is homogeneous, only one refractive-index bin is needed and WP-MS reduces to the angular spectrum method. For an inhomogeneous specimen, the method is more expensive than AS-MS because many FFT-based propagations are required per slice, but it accounts for the local wavelength inside the specimen rather than propagating every slice with the vacuum wavelength.

Because the binning and interpolation are smooth, the update is differentiable with respect to the refractive-index map $n(x, y)$ and can be used in automatic differentiation workflows. The cost is set by the number of bins because each propagation step requires N_{bins} FFT-based propagations rather than one propagation as in AS-MS. This is still much cheaper than evaluating a separate propagation for every distinct pixel value. The accuracy therefore depends on whether the chosen bins resolve the refractive-index distribution of the slice. In practice, we check this by increasing N_{bins} until the slice error reaches the required tolerance.

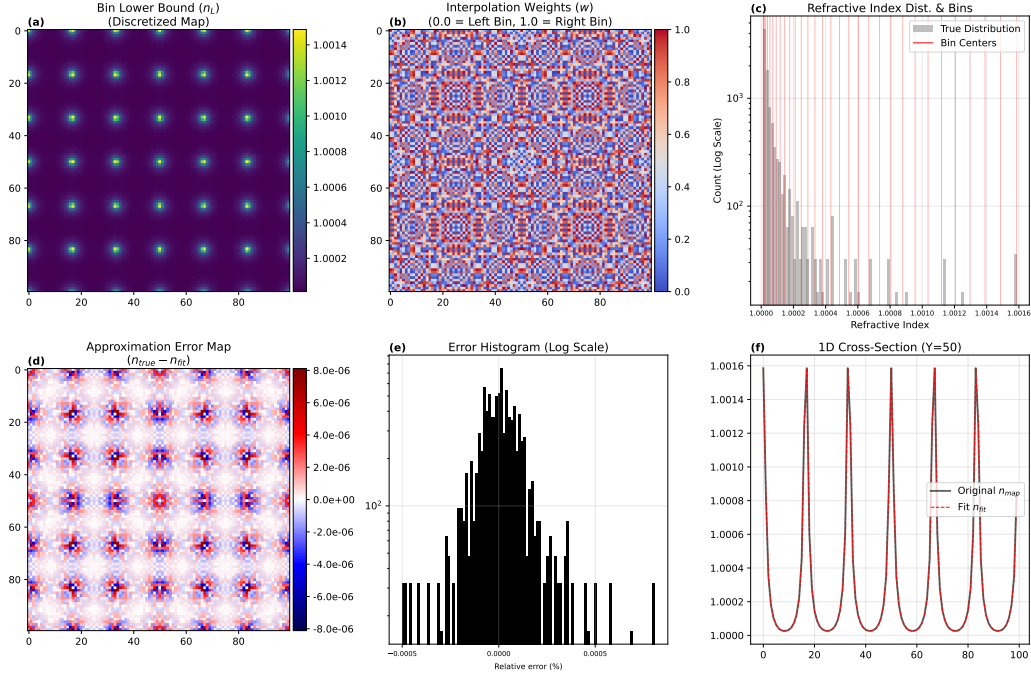


Figure 2: Diagnostics of the differentiable WP-MS binning strategy for a representative Au [100] slice. (a) Lower-bin map n_L , showing the lower refractive index interval inside each pixel. (b) Smooth interpolation weight w , where low values favour the lower-bin propagated field u_L and high values favour the upper-bin propagated field u_R . (c) Histogram of the true refractive-index distribution with the reference-bin positions superimposed. (d) Spatial approximation error $n_{\text{true}} - n_{\text{fit}}$. (e) Histogram of the fitting error on a log scale. (f) One-dimensional cross-section comparing the original refractive-index profile with the interpolated profile n_{fit} .

Figure 2 illustrates how the two-bin interpolation scheme is applied to a representative slice. Panels (a) and (b) separate the binning construction

into the interval assignment and the interpolation coordinate. In panel (a), the lower-bin map identifies the reference interval used at each pixel, whereas panel (b) gives the position of the true refractive index inside that interval through the interpolation weight. The diagnostic refractive-index fit, $n_{\text{fit}} = (1 - w)n_L + wn_R$, uses the same two-bin interpolation as the field update. Panels (c), (e), and (f) show that the chosen non-uniform binning resolves the refractive-index distribution for this slice, while panel (d) shows that the residual interpolation error is spatially concentrated around steep atomic peaks. This supports replacing a separate propagation for every distinct pixel value by a finite bank of propagated fields.

In practice, WP-MS is most useful with GPU acceleration. For the simulations in this work, N_{bins} between 16 and 128 was sufficient for 512×512 and 1024×1024 grids, provided that the binning resolved the refractive-index distribution of the sample.

2.5. Forward ODE reference

We use an ordinary differential equation (ODE) reference obtained by applying a method-of-lines discretisation to the second-order Klein–Gordon Helmholtz equation. The transverse coordinates are represented on the same laterally periodic FFT grid used by the multislice methods, while z is left as the continuous evolution variable. The transverse pixel size and field of view therefore set the retained transverse spatial frequencies, or equivalently the largest scattering angles represented on the grid. The method does not attempt to resolve the rapidly oscillating electron wavelength in all three spatial directions on a Cartesian mesh; instead, the adaptive ODE integration resolves the rapid longitudinal oscillation along z .

Substituting n_{KG} into the Helmholtz equation gives

$$\partial_z^2 \psi = - [\nabla_{\perp}^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)] \psi. \quad (20)$$

Introducing $\phi = \partial_z \psi$ gives the equivalent first-order system

$$\frac{d}{dz} \begin{pmatrix} \psi \\ \phi \end{pmatrix} = \begin{pmatrix} \phi \\ - [\nabla_{\perp}^2 + (2\pi k_0)^2 n_{\text{KG}}^2(x, y, z)] \psi \end{pmatrix}. \quad (21)$$

We integrate this system with an adaptive high-order Runge–Kutta method, using the Dopri8 scheme in the implementation, with a maximum internal step size set from the largest retained discrete frequency. The transverse Laplacian is evaluated spectrally using FFTs, while the n_{KG}^2 term is applied

by pointwise multiplication in real space. The discrete potential supplied to the solver is interpreted as the slice-averaged electrostatic potential, so n_{KG}^2 is constant in z within each slice and discontinuous only at slice boundaries. The adaptive step-size controller is clipped at those boundaries, so no accepted ODE step crosses a jump in the right-hand side. Thus, for a fixed transverse grid and slice stack, the ODE reference solves the corresponding periodic FFT-grid Klein–Gordon problem to the specified ODE tolerance.

To initialise the calculation on the forward-propagating vacuum branch, the entrance wave $\psi(z = 0)$ is supplemented by the longitudinal derivative

$$\phi(z = 0) = \mathcal{F}^{-1} \left[2\pi i \sqrt{k_0^2 - k_\perp^2} \mathcal{F}\{\psi(z = 0)\} \right], \quad (22)$$

where the square-root branch is chosen to give the forward-propagating vacuum solution, and the decaying branch for any evanescent Fourier components on the grid. In our implementation, the same condition is used when no initial $\partial_z \psi$ is supplied; when the ODE solve is restarted slice by slice or cell by cell, the returned exit derivative must be carried forward as the next initial derivative. Otherwise, only ψ is propagated and the second-order state has been changed.

The entrance condition is one-way, but the subsequent evolution is not a one-way propagation approximation. After $z = 0$, the state contains both ψ and $\partial_z \psi$, and the second-order system can generate and carry the backward-propagating component allowed by the discretised initial-value problem. In the Au [100] benchmark, this calculation is applied one unit cell at a time: the ODE solve is restarted for the next repeated cell using both the exit wavefunction and the exit derivative from the previous cell. Thus the complete second-order state is retained across unit cells, rather than being projected back onto the forward vacuum branch at each cell boundary. However, the calculation is still forward initialised at the entrance of the crystal and does not solve a global two-sided scattering problem in which reflections from later depths propagate back into earlier depths. Under the high-energy conditions considered here, we treat the ODE result as a non-paraxial reference, rather than as a global quantitative backscattering calculation. Given the high-energy and weak-potential conditions, the global backward-propagating component is expected to be small, and the ODE reference is an accurate benchmark for the forward-propagating multislice methods.

With this reference defined, the three multislice methods and the ODE benchmark can be seen as different approximations within the same scalar

Helmholtz wave equation framework, once the refractive-index model is fixed. In the ODE benchmark below, all methods use the same per-slice n_{KG} map. F-MS uses a split-step specimen approximation together with the paraxial free-space propagator, and AS-MS keeps the same split-step structure but restores the exact free-space spherical propagator. WP-MS remains one-way, but it does not use a separate transmission step; instead, it incorporates the refractive-index dependence into a local-homogeneous medium propagation update. By contrast, the ODE reference keeps the second-order Klein–Gordon Helmholtz equation after transverse discretisation and evolves the transverse Laplacian and specimen potential simultaneously, which is why it is used as the reference calculation in the first numerical experiment.

3. Benchmarks and Results

We use three numerical experiments to validate our methods against an ODE reference and to benchmark them against each other. The first method reproduces the Au [100] beam-amplitude benchmark of Rother and Scheerschmidt [3], so that F-MS, AS-MS, and WP-MS can be compared with an ODE solution of the full second-order Klein–Gordon equation. The ODE reference is not used in the remaining tests because of its high computational cost. Instead, the second experiment compares the FFT-based methods in a lower- Z Si [111] CBED calculation, and the third uses a thick Au [100] convergent-probe calculation with WP-MS as the practical high-angle reference. This section reports the observed numerical differences; their common interpretation and practical consequences are discussed in Section 4.

3.1. Au [100] ODE Benchmark

To verify the implementation, we recreated the simulation presented in Figure 3 of Rother and Scheerschmidt [3]. The benchmark propagates a plane wave through an Au [100] crystal at 300 keV and compares against a converged ODE solution of the full second-order Klein–Gordon equation. Periodic lateral boundary conditions are implicit because each method uses Fourier propagation, so the crystal is repeated in the transverse directions. To test a thicker object, we extended the original Rother–Scheerschmidt calculation from 25 unit cells (~ 10 nm) to 100 unit cells (~ 40 nm).

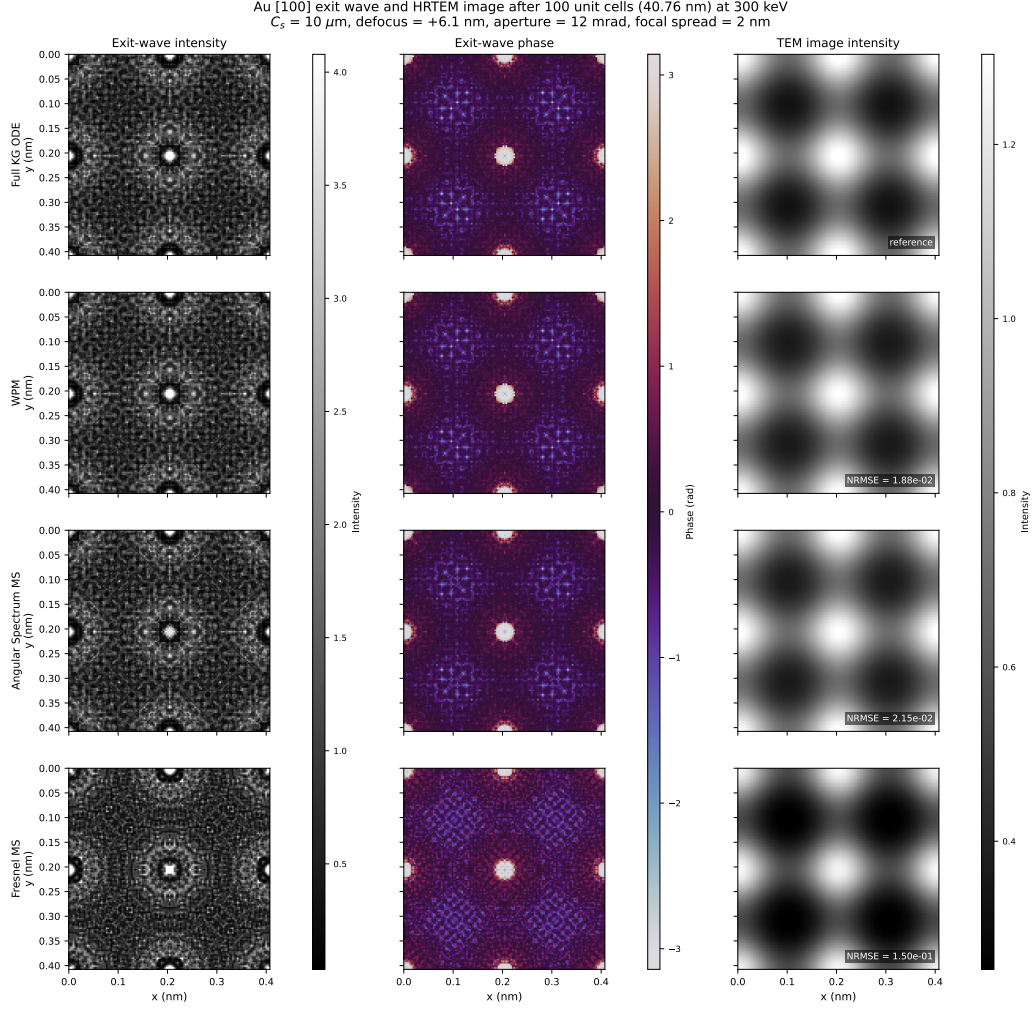


Figure 3: Au [100] ODE verification case after 100 unit cells. The left columns show the real-space exit-wave intensity and phase, and the rightmost column shows the HRTEM image intensity after applying a representative aberration-corrected 300 keV transfer function with residual $C_s = 10 \mu\text{m}$, defocus of +6.0 nm, and a 12 mrad objective aperture. Color scales are shared across methods within each column, and image-plane NRMSE values are relative to the Klein–Gordon ODE reference.

Figure 3 shows the exit-wave intensity, phase, and transfer-function image intensity after propagation through 100 unit cells of Au [100]. The F-MS exit-wave amplitude and phase are qualitatively different from AS-MS, WP-MS, and the Klein–Gordon ODE reference, whereas the latter three remain visually similar. After the microscope transfer function is applied, however,

the image-plane differences are much less obvious because the objective aperture and transfer function suppress part of the high-angle information. The unfiltered exit wave is therefore the more sensitive comparison for this benchmark.

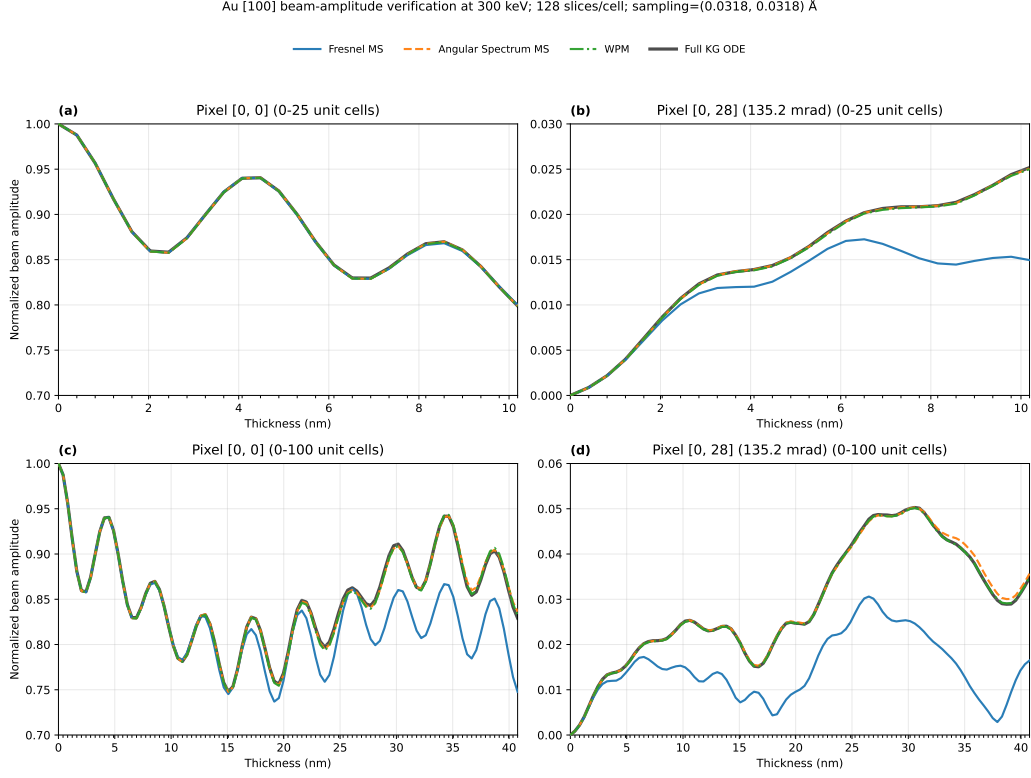


Figure 4: Fourier pixel-amplitude comparison for the Au [100] verification case. The central [0, 0] component and the high-angle [0, 28] component are compared across F-MS, AS-MS, WP-MS, and the Klein–Gordon ODE reference.

Figure 4 gives the corresponding Fourier-amplitude comparison after each slice. For the central [0, 0] component, all methods agree closely up to approximately 10 nm, or 25 unit cells. Beyond this thickness, the F-MS result begins to deviate from the other methods even on the on-axis pixel. By about 15 nm, high-angle diffracted beams have propagated far enough through the crystal to interfere back into the central beam, so an error in their accumulated phase also modifies the on-axis amplitude.

For the high-angle [0, 28] component at approximately 135 mrad, the

F-MS solution diverges from the ODE reference after approximately 10 unit cells, or 4 nm, consistent with the findings of Rother and Scheerschmidt [3]. AS-MS tracks the ODE reference much more closely than F-MS and remains comparable to WP-MS. WP-MS is slightly closer to the reference, but the difference between AS-MS and WP-MS is only clear at the largest thicknesses considered here.

3.2. Si [111] CBED Peak Positions

After verifying AS-MS and WP-MS against the ODE solution, we focus on the three FFT-based multislice methods: F-MS, AS-MS, and WP-MS. The first CBED test uses Si [111] at 100 keV with an 8 mrad probe and a specimen thickness near 1000 Å, following the spirit of the low-energy CBED comparison of Hillier et al. [4] but at a higher accelerating voltage. The maximum scattering angle was set to 300 mrad.

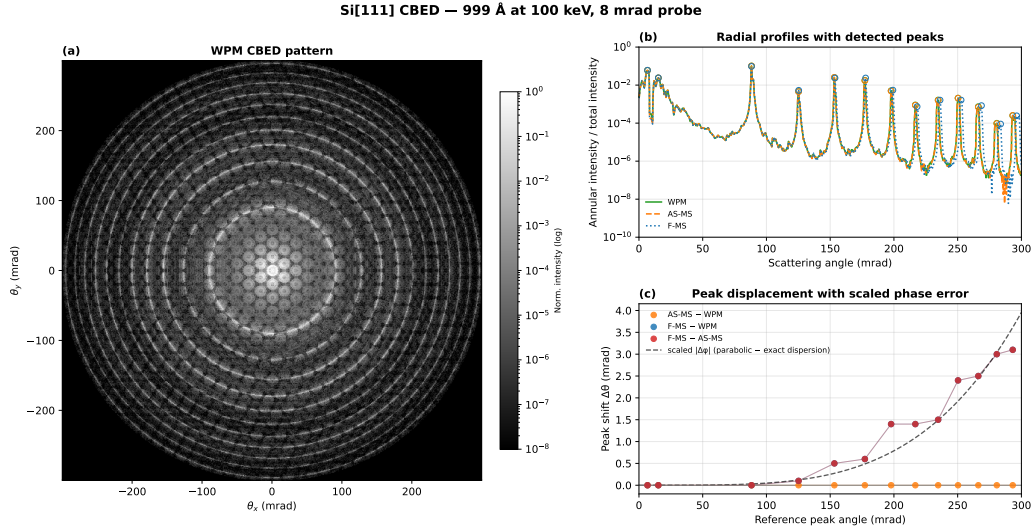


Figure 5: Si [111] CBED comparison at 100 keV, 8 mrad probe, and approximately 1000 Å thickness. (a) WP-MS CBED pattern cropped to 300 mrad. (b) Radially integrated annular intensity for all three methods, with detected diffraction peaks marked. (c) Peak displacement relative to WP-MS, with a scaled parabolic-versus-exact dispersion error curve overlaid. The F-MS—WP-MS shift (blue) follows the dispersion-error curve closely, while the AS-MS—WP-MS shift (orange) remains near zero, consistent with the Au [100] results.

Figure 5 shows the WP-MS CBED pattern, the radial profiles with detected peaks, and the peak displacements relative to WP-MS. The F-MS

peaks shift progressively with scattering angle. By contrast, AS-MS tracks WP-MS closely, with peak displacements near zero across the full 300 mrad range.

In Figure 5(c), the detected peak shifts are compared with a rescaled dispersion-relation phase error, namely the difference between the parabolic Fresnel dispersion and the exact spherical dispersion,

$$|\Delta\phi| = \frac{2\pi}{\lambda} \left| \sqrt{1 - \lambda^2 q^2} - 1 + \frac{1}{2} \lambda^2 q^2 \right|,$$

where $q = \sin \theta / \lambda$ is the transverse spatial frequency. This is the longitudinal phase error introduced when the exact spherical propagator is replaced by its Fresnel approximation. After fitting only the overall scale, the curve follows the F-MS – WP-MS peak shifts closely, indicating that the observed displacement is mainly caused by accumulated free-space propagation error between slices. In this Si case, the additional WP-MS correction from the potential-dependent wavelength inside the specimen is not clearly resolved in the peak positions.

3.3. Au [100] CBED Amplitude Errors

To test whether a higher atomic number increases the differences between the FFT-based methods, we simulated convergent-beam electron diffraction from an Au [100] supercell at 200 keV using a 5 mrad aberration-free probe. The probe was propagated to a specimen thickness of approximately 100 nm. To sample the high-angle components of the convergent beam accurately, the enlarged 9×9 gold cell was discretized on a 2048×2048 grid. We then propagated the wave using F-MS, AS-MS, and WP-MS, and compared the exit-surface CBED patterns obtained from the Fourier transform of each exit wave.

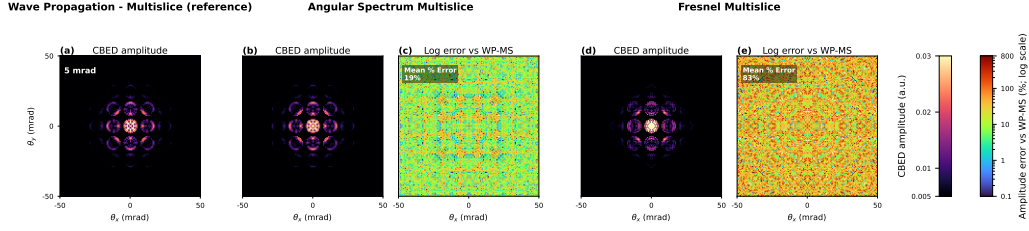


Figure 6: CBED exit-wave amplitude comparison for the Au [100] convergence-angle experiment. The maps show the final diffraction patterns produced by F-MS, AS-MS, and WP-MS, together with log-scale amplitude errors relative to WP-MS.

Figure 6 compares the final diffraction-plane amplitudes. The F-MS CBED amplitude differs visibly from WP-MS, especially in the higher-angle features of the pattern. AS-MS remains much closer to WP-MS, and the remaining AS-MS–WP-MS amplitude difference is small compared with the F-MS–WP-MS discrepancy.

Au [100] CBED amplitude relative RMSE vs thickness (reference: WP-MS), 200 keV, 64 slices/cell

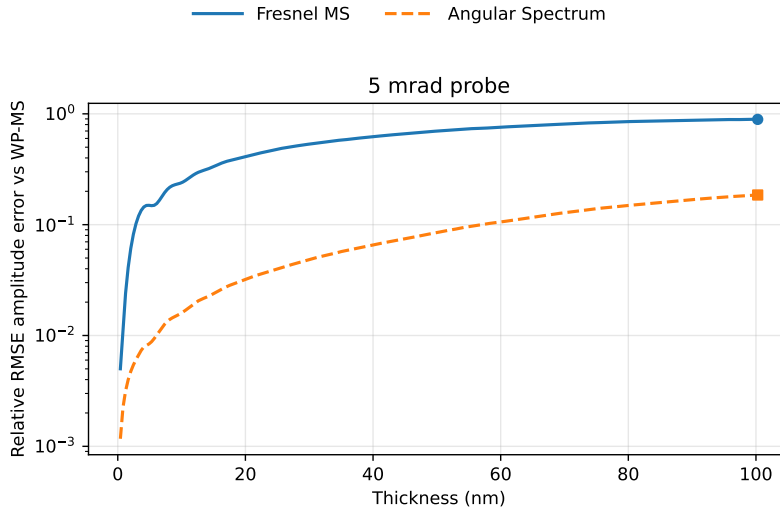


Figure 7: Relative CBED amplitude error as a function of specimen thickness for the Au [100] convergence-angle study. The error is reported against the WP-MS reference at the end of each unit cell.

Figure 7 shows the root mean square error of the CBED amplitude against the WP-MS reference as a function of thickness. After approximately 10 nm, the F-MS amplitude error is about an order of magnitude larger than the AS-MS error. The AS-MS error increases with thickness, but it remains much closer to WP-MS than to F-MS throughout the calculation. A residual AS-MS–WP-MS difference is also visible in the Au [100] exit-wave comparison of Figure 3, even at 40 nm.

4. Discussion

The three benchmarks point to a common source of error in conventional multislice under the high-angle, thick-specimen conditions tested here. In the Au [100] ODE benchmark, the F-MS discrepancy appears first in the 135 mrad beam and only later in the central beam, after diffracted components have propagated far enough through the crystal to interfere back into it. In the Si [111] CBED calculation, the F-MS peak shifts follow the parabolic-versus-exact dispersion error, while AS-MS remains close to WP-MS over the measured angular range. In the thick Au [100] CBED calculation, the F-MS amplitude error becomes roughly an order of magnitude larger than the AS-MS error beyond about 10 nm. Taken together, these observations indicate that the dominant F-MS error in these tests is the accumulated longitudinal phase error from the parabolic Fresnel propagator.

4.1. AS-MS as the minimal wide-angle correction

AS-MS removes most of this error by changing only the free-space propagation kernel. The transmission step, split-step ordering, FFT structure, and number of FFTs per slice are the same as in F-MS. In the spatial-frequency convention used here, the replacement is

$$\exp(-i\pi k_{\perp}^2 \Delta z / k_0) \rightarrow \exp\left(2\pi i \Delta z [\sqrt{k_0^2 - k_{\perp}^2} - k_0]\right).$$

The correction is small when all important Fourier components satisfy $k_{\perp} \ll k_0$, but it becomes significant when the simulation retains components with large $k_{\perp}/k_0 = \sin\theta$. Once those components are present, the error is not confined to high-angle beams because dynamical scattering can later mix them back into lower-angle amplitudes.

This is the main practical result of the comparison. If a simulation grid already represents the high-angle scattering of interest, then replacing the

Fresnel kernel by the angular-spectrum kernel removes the dominant wide-angle propagation error in these benchmarks without changing the basic multislice workflow. Under those conditions, there is little practical reason to retain the Fresnel propagator except for direct comparison with legacy calculations or for regimes where the paraxial error is already negligible relative to other uncertainties.

4.2. Sampling and cost

The practical advantage of AS-MS would be less useful if the exact angular-spectrum phase required a substantially finer grid than F-MS. The additional sampling concern comes from the angular-spectrum propagation phase, which is steeper than the Fresnel phase at large transverse wavevector. For one propagation step,

$$\phi_{\text{F}}(k_{\perp}) = -\pi \frac{\Delta z}{k_0} k_{\perp}^2, \quad \phi_{\text{ASM}}(k_{\perp}) = 2\pi \Delta z \left[\sqrt{k_0^2 - k_{\perp}^2} - k_0 \right].$$

Comparing the phase gradients gives the relative axial sampling penalty

$$\frac{|d\phi_{\text{ASM}}/dk_{\perp}|}{|d\phi_{\text{F}}/dk_{\perp}|} = \frac{k_0}{\sqrt{k_0^2 - k_{\perp}^2}}.$$

This factor is only large when $k_{\perp}$ approaches k_0 , where the propagating spectrum approaches the evanescent boundary. It can be written directly in terms of scattering angle because $k_{\perp}/k_0 = \sin \theta$:

$$\frac{k_0}{\sqrt{k_0^2 - k_{\perp}^2}} = \frac{1}{\sqrt{1 - \sin^2 \theta}} = \frac{1}{\cos \theta}.$$

Under the TEM conditions used here this remains close to unity. For a 200 mrad probe, the largest correction to the Fresnel slice-thickness condition is

$$\frac{1}{\cos(200 \text{ mrad})} \approx 1.02,$$

so the axial sampling bound is tightened by only about 2%. The transverse grid sets a separate limit because it determines the largest scattering angle that can be represented without aliasing. A real-space pixel size

Δx gives a maximum transverse spatial frequency $k_{\perp, \text{Nyq}} = 1/(2\Delta x)$. Since $k_{\perp} = k_0 \sin \theta$ and $k_0 = 1/\lambda$, the corresponding Nyquist angle is

$$\theta_{\text{Nyq}} = \arcsin\left(\frac{\lambda}{2\Delta x}\right) \approx 320 \text{ mrad}.$$

Here we used $\lambda \approx 0.025 \text{ \AA}$ at 200 keV and $\Delta x = 0.04 \text{ \AA}$. This angle is still far from the evanescent boundary at $\theta = 90^\circ$, where the square root in the ASM kernel becomes singular. In the present benchmarks, AS-MS therefore has the same per-slice FFT cost as F-MS and does not require substantially finer axial or transverse sampling once the simulation already resolves the high-angle features of interest.

4.3. WP-MS contribution

WP-MS makes a further correction by allowing the propagation kernel to depend on the local electrostatic refractive index, so the wavelength changes inside each slice. In the electron-optics setting, this gives a differentiable one-way local-medium propagation update that retains FFT-based propagation while incorporating a potential-dependent wavelength.

WP-MS is closest to the ODE reference in the Au [100] benchmark, and the residual AS-MS–WP-MS difference in the thick Au [100] CBED calculation is consistent with this additional local-wavelength correction. The refractive index near the centre of a gold atom is approximately 1.07 in this simulation, large enough that the correction can accumulate over tens of nanometres. Under the high-energy conditions considered here, however, this effect is smaller than the improvement obtained by replacing the Fresnel propagator with the angular-spectrum propagator. In the high-energy context tested here, WP-MS provides little extra accuracy for its additional computational cost, while AS-MS provides most of the practical correction at essentially the same per-slice cost as F-MS.

4.4. Consequences and limits

The silicon CBED calculation (Fig. 5) shows that the same propagator error is not limited to the most severe high- Z case. Silicon scatters more weakly than gold, so the CBED amplitude differences are smaller than in the thick Au [100] calculation, but the F-MS peak shifts still follow the angular dispersion error. These high-angle peaks would often be damped by thermal diffuse scattering in a room-temperature TEM experiment, particularly beyond about 200 mrad, and a conventional image simulation can hide

part of the error if the transfer function or objective aperture suppresses the affected Fourier components. CBED and ptychographic forward models use diffraction-plane intensities more directly, so the same propagator error can become more visible.

In a ptychographic reconstruction, the forward model predicts detector intensities for each probe position. If the propagator gives the wrong high-angle amplitudes, peak positions, or accumulated phase through a thick specimen, the optimization can compensate through the recovered potential. Although ptychographic reconstruction for specimens as thick as the Au [100] CBED test is not yet routine, the calculation shows that the choice of propagator can change the diffraction-plane amplitudes before other experimental effects are added. For forward models that already use F-MS, most of the wide-angle correction can therefore be recovered by replacing the Fresnel kernel with the angular-spectrum kernel, while keeping the same transmission step and per-slice FFT complexity.

The comparison here remains restricted to high-energy, elastic, forward-scattering calculations. Lower accelerating voltages should be tested separately because wide-angle propagation, potential-dependent wavelength changes, and backscattering effects become more significant at lower energies [4–6]. The WP-MS implementation used here is one-way, but unlike the AS-MS or F-MS methods, can be formulated into a form that can include bidirectional propagation [25–27]. Adapting the bidirectional WP-MS to electron optics could then provide a practical alternative to the FC-MS method of Chen and Van Dyck [8].

5. Conclusion

We adapted the optimized wave propagation method developed in scalar optics to electron microscopy, giving a wave-propagation multislice (WP-MS) method that tracks high-angle propagation and local wavelength changes within a differentiable slice-based propagation framework. We then compared F-MS, AS-MS, and WP-MS in three complementary settings: an Au [100] ODE benchmark, a lower- Z Si [111] CBED peak-position test, and a thick Au [100] CBED amplitude test.

Across these tests, replacing the Fresnel kernel with the exact angular-spectrum kernel accounts for most of the improvement over conventional multislice. AS-MS keeps the same transmission step, FFT structure, and per-slice FFT cost as F-MS, and under the conditions studied here it does not

require substantially finer sampling once the high-angle features are already represented. For high-angle elastic forward simulations where F-MS is still being used, AS-MS is therefore the most direct practical default.

WP-MS is the more complete one-way local-medium propagation model and is closest to the ODE reference in the Au [100] benchmark, but its additional accuracy over AS-MS is modest under the high-energy conditions tested here. Its main role in this work is to bring the wave propagation method from scalar optics into electron optics and to provide a differentiable local-wavelength extension of angular-spectrum multislice. Future work should test the performance of WP-MS at lower accelerating voltages, against the bidirectional FC-MS method and inside ptychographic reconstructions where systematic forward-model errors could bias the recovered potential.

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